Lab 1: Requirements Engineering & UML Use-Case Modelling

Name: Vankadara Sri Kruthi
SRN: PES2UG24CS574
Subject: Software Engineering
Section : 5J
Scenario : Corporate Mental Wellness & Counselling Platform (PROBLEM STATEMENT #14)

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate an anonymized pseudo-ID for employee accounts to book counseling sessions without revealing the corporate email or employee ID to the therapist.	High	Pass: Therapist dashboard displays pseudo-ID only. Fail: Employee PII exposed in booking metadata.	Confidentiality of employee identity is the core value proposition of the platform and is required for employees to trust and adopt the service.
FR-002	Functional	The system shall present employees with a validated stress-assessment questionnaire (e.g., a 10-item PSS-style survey) and compute a stress score (0-40) immediately after submission.	High	Pass: A numeric score in the range 0-40 is displayed within 2 seconds of submission and saved to the employee's confidential wellness history. Fail: No score is generated, or the score is visible to any role other than the employee.	Self-assessment is the entry point for employees to gauge their wellbeing and decide whether to book a session.
FR-003	Functional	The system shall allow an employee to view a licensed therapist's available time slots, filtered by specialization and language, and confirm a booking using only the employee's pseudo-ID.	High	Pass: A confirmed booking appears on both the employee's and therapist's calendars with matching pseudo-ID and time slot, and no double-booking occurs. Fail: A session is booked into an unavailable slot or the employee's real identity appears on the therapist's calendar.	Booking is the primary functional goal of the platform; accurate scheduling and identity protection are both essential.
FR-004	Functional	The system shall allow employees to log daily wellness indicators (mood, sleep hours, stress level) and shall display a rolling 30-day trend chart of these indicators to the employee.	Medium	Pass: The trend chart updates within one entry cycle and every logged data point is reflected accurately (0% data loss). Fail: The chart omits entries or values differ from logged data by more than 5%.	Longitudinal self-tracking helps employees and, with consent, their therapist monitor progress between sessions.
FR-005	Functional	The system shall allow an employee to cancel or reschedule a booked session up to 4 hours before the scheduled start time without requiring a reason.	Medium	Pass: A cancellation/reschedule request made more than 4 hours before the session succeeds and immediately frees the slot for other employees. Fail: A request	Provides employees flexibility to manage their schedules while protecting therapists from last-minute cancellations.

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				inside the 4-hour window is accepted without a late-change flag, or a valid request outside the window is rejected.	
NFR-001	Nonfunctional	The appointment scheduler must handle timezone conversions accurately across global office locations.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load, with all displayed appointment times correctly converted to each employee's local timezone. Fail: Any appointment time is displayed in the wrong timezone, or latency/security benchmarks are not met.	The corporate workforce is distributed across multiple offices and timezones; incorrect scheduling would directly cause missed sessions.
NFR-002	Nonfunctional	The system shall encrypt all pseudo-ID-to-employee mapping tables and session records using AES-256 at rest and TLS 1.3 in transit, and shall never write employee-identifying data to application logs.	High	Pass: A security audit confirms AES-256 encryption at rest and TLS 1.3 for all data channels, with no plaintext PII found in logs or backups. Fail: Any unencrypted storage or transmission of identifying data is detected.	Protects employee confidentiality, the platform's core value proposition, and supports compliance with data-protection regulations (e.g., HIPAA-aligned, GDPR/DPDP).

Actors and Use Cases:

Actors:

1. Corporate Employee
2. Licensed Therapist
3. HR Wellness Analyst
4. Notification/Reminder Service *(system actor, if you want a 4th)*

Use Cases:

Assess Stress Level

- Employee completes a stress-assessment questionnaire and receives a computed stress score.

Generate Anonymized Pseudo-ID

- System issues/confirms a pseudo-ID so the employee's real identity is never shown to the therapist. *(always invoked — «include» of Book Counseling Session)*

Book Counseling Session

- Employee views a licensed therapist's availability and confirms a session using their pseudo-ID.

Trigger Crisis Escalation Alert

- System flags and escalates the case if the stress-assessment score crosses a critical threshold. *(optional — «extend» of Assess Stress Level)*

Track Wellness Indicators

- Employee logs daily mood/sleep/stress data and views a trend chart over time.

Cancel / Reschedule Session

- Employee cancels or reschedules a booked session before the cutoff window.

Manage Session Notes

- Licensed Therapist records confidential session notes tied to the pseudo-ID.

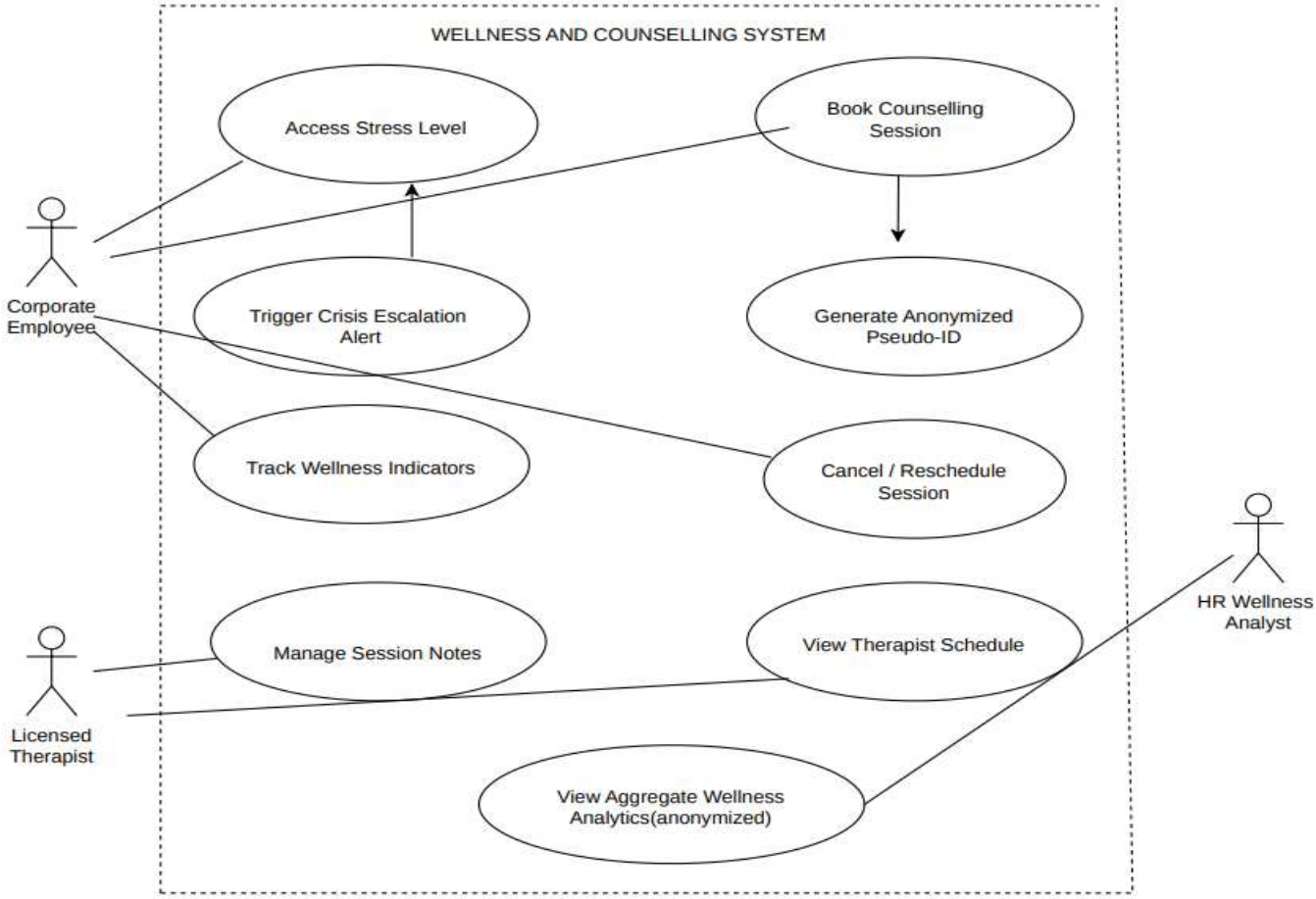
View Therapist Schedule

- Licensed Therapist views upcoming confirmed sessions.

View Aggregate Wellness Analytics

- HR Wellness Analyst views only anonymized, aggregated wellness trends — never individual employee data.

UML Use-Case Diagram:



- **Use Case Name:** Submit and Confirm Counseling Session Booking
- **Primary Actor:** Corporate Employee — the one who initiates the use case
- **Supporting Actors:** Licensed Therapist (whose calendar/availability the system checks and who receives the anonymized booking), and the System's Pseudo-ID Service (invoked via the «include» relationship to anonymize the employee)

Preconditions:

- The employee has a valid corporate account and has been issued (or already holds) an anonymized pseudo-ID for the platform.
- The employee is authenticated and logged in to the counseling portal.
- At least one licensed therapist has published available time slots that match the employee's preferred specialization and language.

Postconditions:

- A confirmed counseling session exists on both the employee's and the therapist's calendars, linked only by the employee's pseudo-ID.
- A confirmation notification (with session time, mode, and reschedule/cancel link) is sent to the employee.
- No corporate email, employee ID, or other identifying information is stored in, or visible from, the therapist's booking record.

Main Success Scenario:

1. Employee selects "Book Counseling Session" from the portal's home screen.
2. System invokes (Generate Anonymized Pseudo-ID) to confirm or issue the employee's pseudo-ID for this session («include»).
3. System displays a list of licensed therapists filtered by specialization, language, and time-zone-adjusted availability.
4. Employee selects a therapist and a preferred time slot from the displayed availability.
5. System locks the selected slot and prompts the employee to confirm the session (date, time, mode: video/audio/chat).
6. Employee confirms the booking.
7. System creates the session record using the pseudo-ID only, updates both calendars, and releases the lock on any other conflicting slot.
8. System sends a confirmation notification to the employee and an anonymized booking notice to the therapist.
9. Use case ends successfully.

Alternate Flow:

5a. Selected Slot No Longer Available

5a1. System detects that another employee booked the same slot before confirmation.

5a2. System displays a "slot no longer available" message and refreshes the therapist's availability list.

5a3. Employee selects a different available time slot.

5a4. Flow resumes at Main Success Scenario step 5.